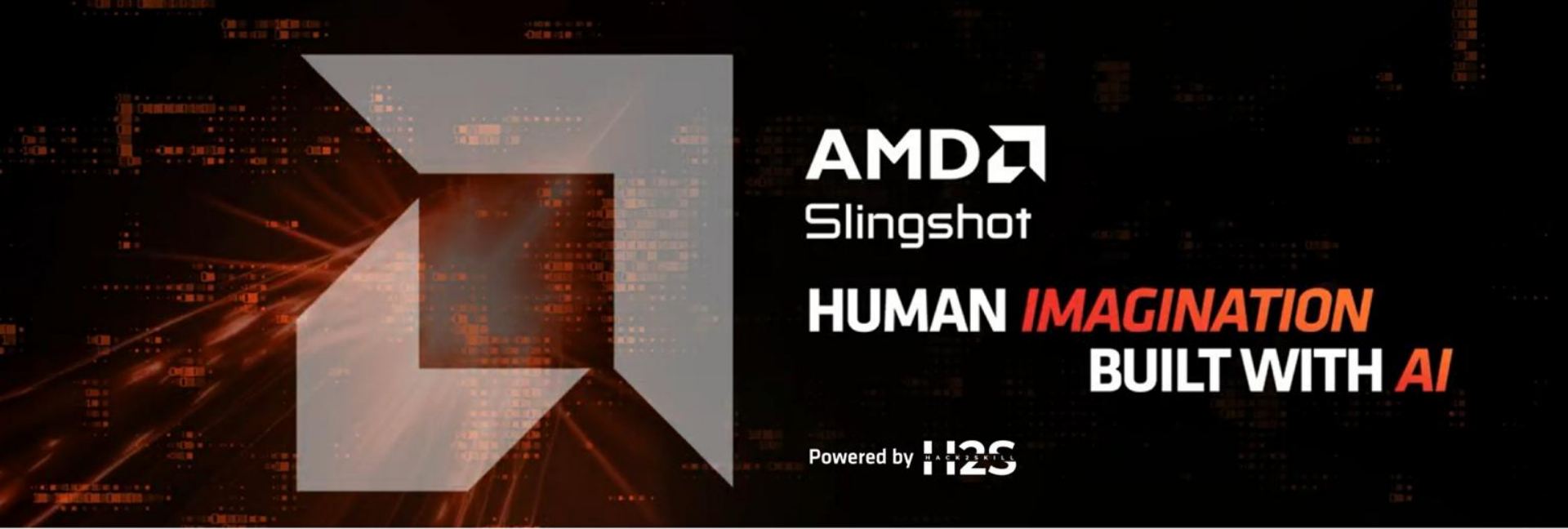




Team Details

- **Team name** – InnoMinds
- **Team leader name** – Chagalamarri Ruthvika
- **Problem Statement** -Rural elderly people lack a simple, offline, voice-based assistant in local languages to manage daily well-being, medicines, fraud risks, government scheme information, access nearby hospitals, banks and government offices, and to read/explain stored books and play familiar songs without internet or cloud dependence

Brief about the idea

- Introducing **ElderConnect**
- A Standalone, offline, voice-based AI for rural elderly, running fully on a low-cost Android smartphone.
- Speaks and understands local languages & elderly speech patterns; no reading, typing, or English required.
- Supports daily well-being check-ins and medicine guidance with voice reminders.
- Sends alerts to trusted contacts only when the elder says they are not feeling well.
- Provides fraud & scam warnings for unknown calls and OTP/money requests.
- Guides elders to nearby hospitals, PHCs, banks, Panchayat offices, ration shops, and essential services using spoken directions.
- Explains government schemes and updates from a locally stored database that updates automatically when internet is available.
- Reads and explains pre-downloaded books and plays familiar songs offline for Companionship and all processing is on-device, no cloud or data sharing.



Why ElderConnect is Needed??

- Popular voice assistants like Google Assistant and Alexa rely heavily on continuous internet connectivity, which is unreliable in many rural areas.
- These assistants are general-purpose tools, not designed specifically for the daily needs, safety, and comfort of elderly users.
- They often assume English literacy, reading ability, and technical familiarity, creating barriers for many elders.
- ElderConnect is purpose-built for elderly care—offline, voice-only, local-language, and privacy-first.
- It brings together health support, medicine reminders, fraud protection, service navigation, and companionship in a single, simple solution.



How different is it from any of the other existing ideas?

Existing System

- Require continuous internet connectivity
- Mostly work in English or limited formal languages
- Depend on cloud servers, raising privacy concerns
- Designed for tech-savvy users, not elderly
- Focus on single use cases (only reminders, only calling)
- Provide generic responses, not personalized to village life
- No protection against fraud and scam calls
- Do not support offline reading, explanation, or companionship



Proposed System

- Works completely offline on a low-cost Android phone
- Communicates in local languages & rural dialects
- All processing happens on-device → zero cloud dependency
- Designed specifically for elderly users .
- Combines health support, safety, service guidance, fraud prevention, and companionship in one app.
- Uses spoken, landmark-based directions for rural areas.
- Warns elders about scam calls and OTP fraud.
- Can read & explain stored books and play familiar songs offline
- Alerts family only when the elder explicitly says they are not feeling well



How will it be able to solve the problem?

Works without Internet

Runs fully offline on a low-cost Android phone, making it reliable in rural areas with poor connectivity.

Accessible to All Elders

Uses voice interaction in local languages and elderly-friendly speech, removing the need for reading, or typing

Supports Health & Daily Well-Being

Provides medicine guidance, voice reminders, and daily check-ins to help elders manage their routine independently.

✨ ElderConnect solves the problem by making essential care, safety, and services accessible through offline voice interaction.

Safe & Trust-Based Alerts

Notifies trusted contacts only when the elder says they are not feeling well, respecting autonomy and avoiding false alarms.

Protects from Fraud & Simplifies Services

Warns against scam calls/OTP fraud and guides to nearby essential services using spoken directions.

Private & Emotionally Supportive

Explains government schemes from offline data, reads stored books, plays familiar songs, and no cloud sharing.

USP of the proposed solution

Truly Offline & Standalone AI

Works fully offline on low-cost Android phones, ideal for rural areas with poor or no internet.

Designed Exclusively for Rural Elderly Users

Voice-only interaction in local languages—no reading, typing, or English needed.

Privacy-First, On-Device Intelligence

All processing happens on-device with no cloud or data sharing.

Trust-Based Alert Mechanism

Alerts are sent only when the elder explicitly asks for help.

All-in-One Daily Life Companion

Combines care, safety, services, and companionship in one app.

Fraud & Scam Protection for Vulnerable Elders

Actively warns elders against scam calls and OTP/money fraud.

Offline Access to Essential Services Information

Guides elders on government schemes and nearby essential services.

Emotional & Social Support Without Internet

Reads books and plays familiar songs offline for companionship.

List of features offered by the solution



Offline, Standalone System



Voice-Based AI Assistant



Local Language Support



Medicine Guidance & Reminders



Daily Well-Being Check-Ins



Trust-Based Family Alerts



Fraud & Scam Warning



Nearby Essential Services

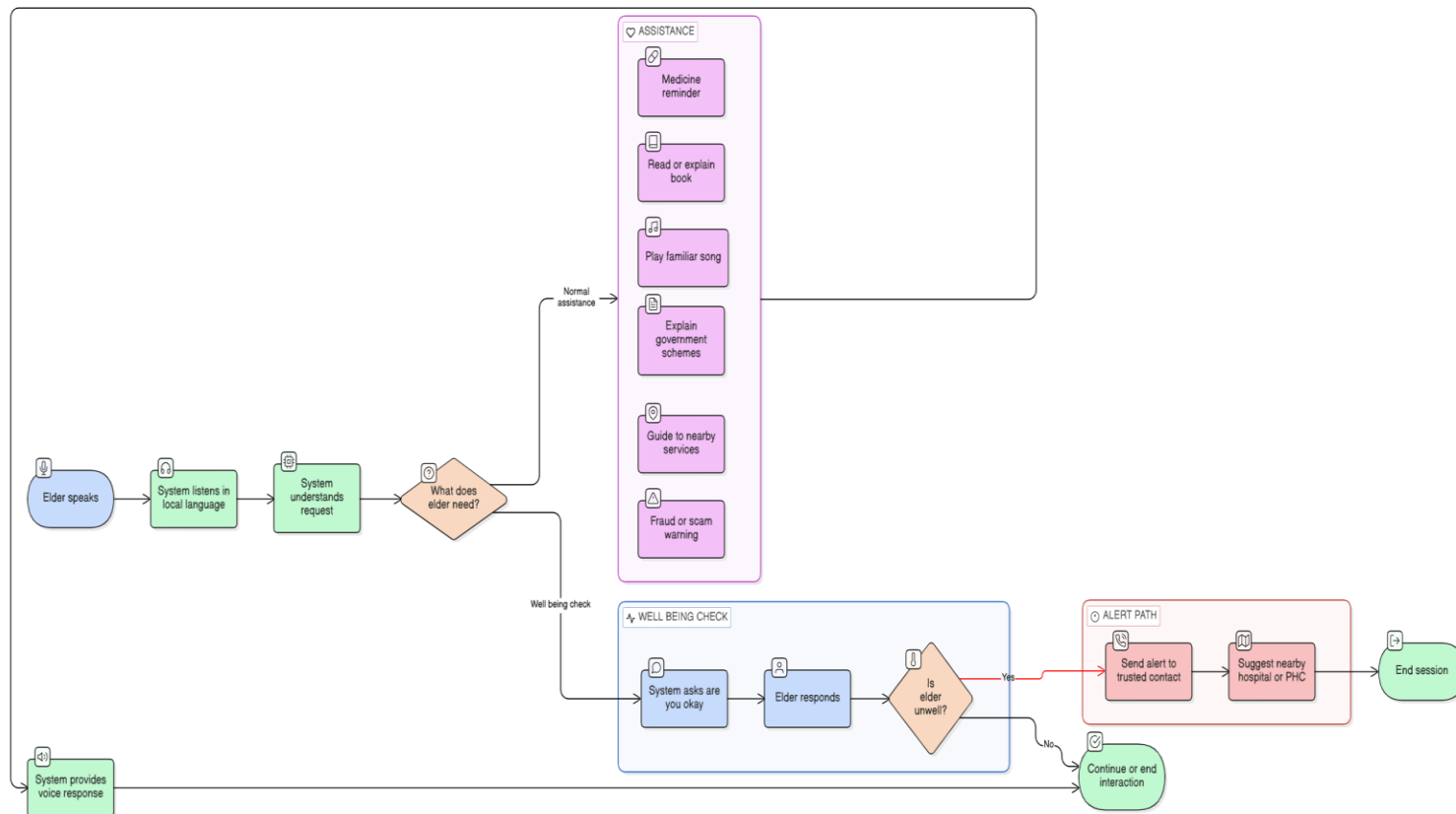


Government Schemes Information

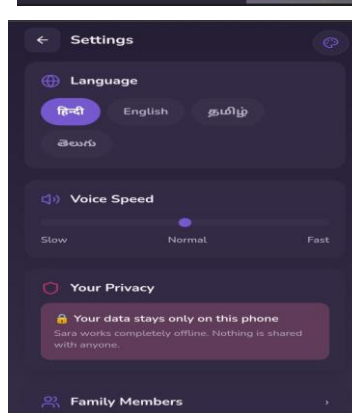
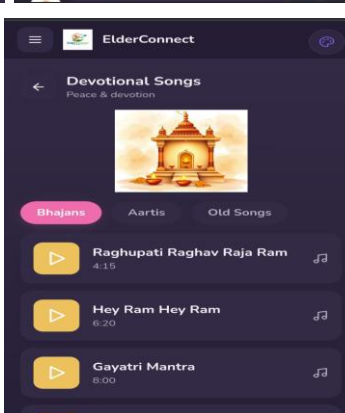
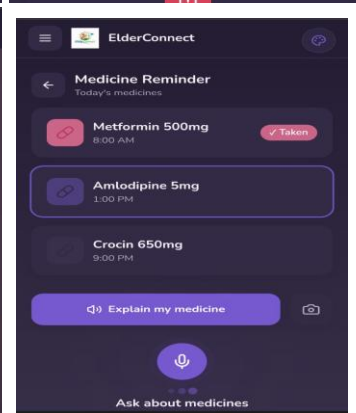
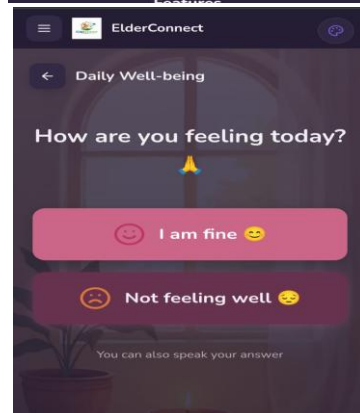
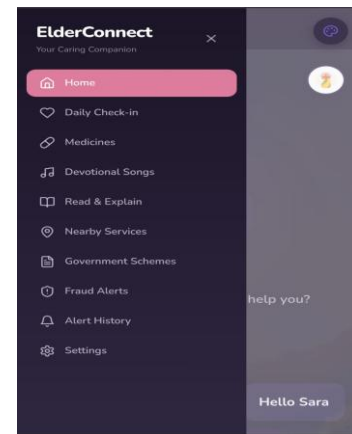
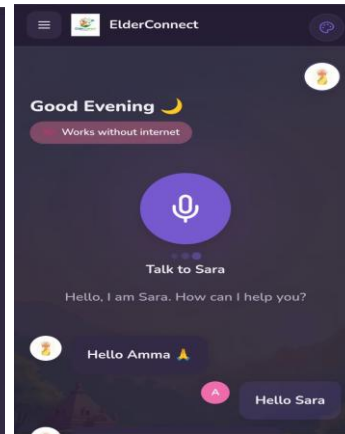
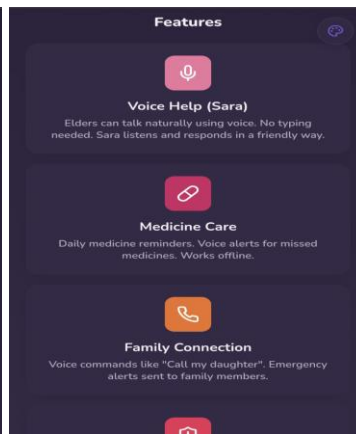
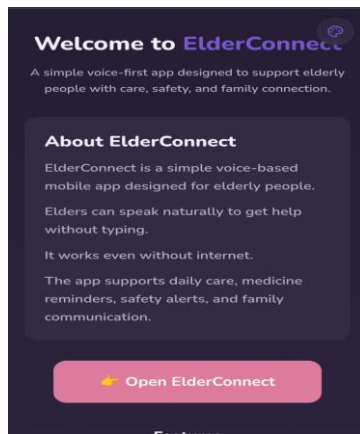
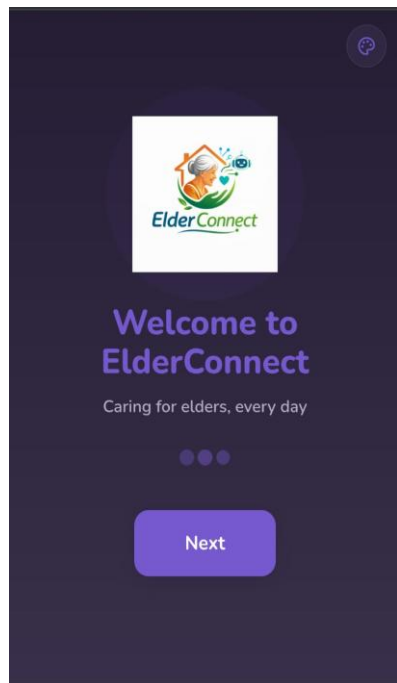


Read ,Explain Books, Play Songs & Companionship

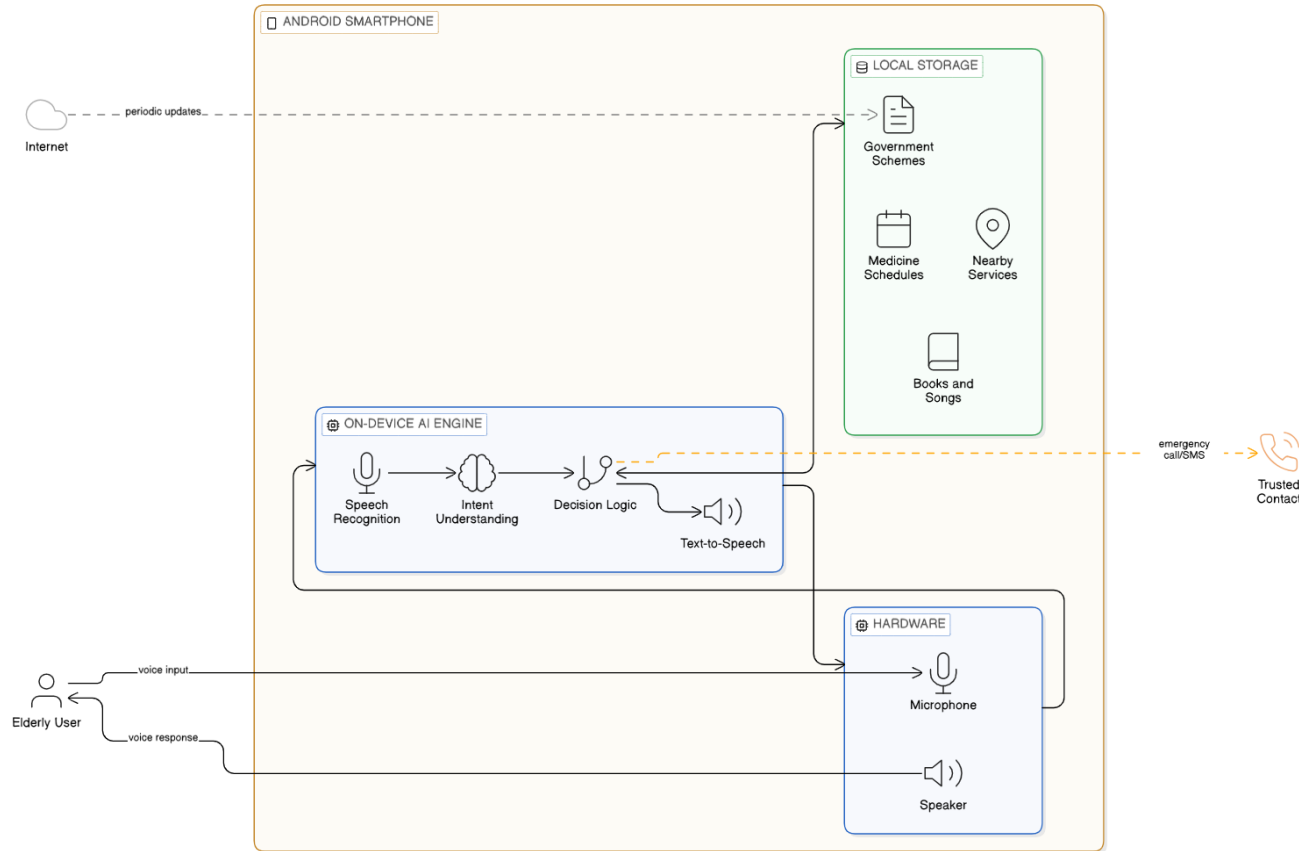
Process flow diagram



Mock diagrams of the proposed solution



Architecture diagram of the proposed solution



Technologies to be used in the solution



Android OS

Low-cost smartphones, offline support



Frontend

- Android UI
- Material Design
- Accessibility
- Animations



Voice Interaction

- Offline Speech Recognition
- Text-to-Speech
- Microphone input



On-Device Backend

- Natural Language Understanding
- Rule-based decision engine
- Context management



Safety & Privacy

- Fraud detection logic
- Voice warnings
- On-device processing
- No cloud, permission-based access



Communication

- Android Telephony API (calls)
- SMS Manager API (messages)



GPS & Geolocation (Offline Mapping)

For providing landmark-based directions to nearby essential services.



Local Storage

- SQLite (medicines, contacts, schemes)
- Local files (books, songs)

These technologies are carefully selected to ensure the solution is robust, private, and highly effective within the rural elderly context.

Usage of AMD Products/Solutions

AMD Ryzen™ Processors

Used for:
Android app development
UI/UX design testing
Offline AI logic simulation
Voice interaction testing

AMD Edge AI Alignment

Used for:
Running AI logic locally on
the device
Avoiding cloud dependency
Reducing latency and
improving privacy

Performance & Efficiency

Used for:
Efficient handling of voice
processing workflows
Fast local data processing
Low-latency response
during testing and
simulation

Scalability with AMD Ecosystem

Suitable for rural, low-
connectivity environments
where edge computing is
critical.

“The project leverages AMD-powered systems for development and aligns with AMD’s Edge AI approach by enabling offline, on-device AI processing.”

Prototype Assets

Demo Video Link

<https://www.instagram.com/reel/DVWGCqOAfnH/?igsh=cnM4OGtrcXRuaThn>



AMD
Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by **I I2S**

Thank you!

